

Lab Class: Physics

Determination of the mass moment of inertia of a wheel

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1 Introduction

This lab experiment demonstrates the behavior of a rigid body in an oscillation. It is part of a university course and serves as a learning tool. It specifically has the goal to examine the relationships between the moment of inertia and the properties of an oscillation.

2 Theory

The mass moment of inertia (I) is a measure of an object's resistance to angular acceleration about a given axis. For a solid cylinder with radius (R) rotating about its symmetry axis, it is defined as:

$$I = \frac{1}{2} \cdot m \cdot R^2 \quad (1)$$

If the axis of rotation is displaced by a distance s from the center of mass, the moment of inertia I' about the new axis is given by:

$$I' = I + m \cdot s^2 \quad (2)$$

When a disk or wheel is suspended from a wire and twisted, it undergoes **torsional oscillation** due to the restoring torque from the wire. In this case the moment of inertia can be determined using the angular directrix of the wire (D^*) and the period of the oscillation (T). It is given by:

$$I = D^* \cdot \left(\frac{T}{2\pi}\right)^2 \quad (3)$$

Therefore, the relationship between the moment of inertia of a disk (I_d) and wheel (I_w) suspended on the same wire with corresponding periods T_d and T_w can be expressed as:

$$I_w = I_d \cdot \frac{T_w^2}{T_d^2} \quad (4)$$

If a wheel is suspended as a **physical pendulum**, it oscillates under gravity. The moment of inertia (I'_w) with respect to that axis of rotation is given by:

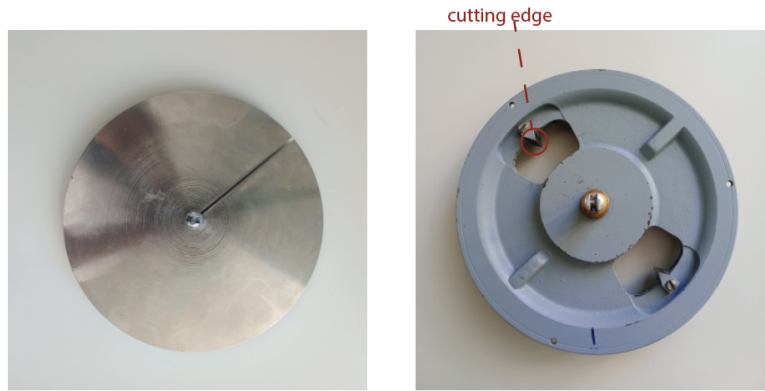
$$I'_w = m \cdot g \cdot s \cdot \left(\frac{T_w}{2\pi}\right)^2 \quad (5)$$

where, g is gravitational acceleration, s is the distance of the axis from the center of mass and T_w is the period of the oscillation

3 Experimental Setup

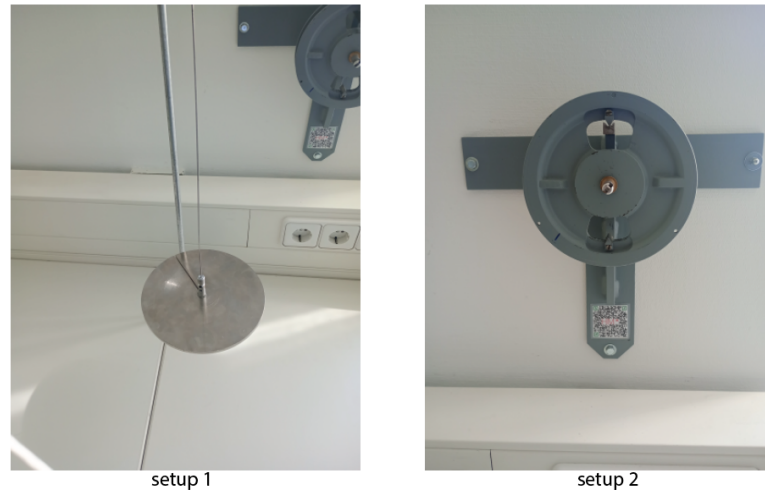
The experiment was set up in a laboratory provided by the *Hochschule München University of Applied Sciences*, which is used for a variety of experiments on classical mechanics (room, G247).

This experiment used two different pendulums. First a wire with a removable disc ($m_d = 1158 \pm 0.05g$, $d_d = 178 \pm 0.3mm$) suspended in the middle as a torsion pendulum. The disc serves as a control object for measurements to compare to the measurements of a wheel ($m_w = 1784.3 \pm 0.05g$, $d_w = 102.95 \pm 0.05mm$), the actual subject of the experiment. The same wheel was used for the second pendulum, a physical pendulum.



(6)

For the second part, the wheel was suspended as a physical pendulum from the cutting edge of the wheel.



(7)

The radius of the wheel and the disk were measured using a vernier caliper with an uncertainty of $0.05mm$, and the masses were measured using a counting scale with an uncertainty of $0.01g$.

4 Experimental Produce

The measurements were executed using the two setups as follows:

a. Torsion Pendulum

For this setup, the disk was first suspended freely from the wire and was rotated 180° before it was released so it starts to oscillate. After, the time t_d of 15 oscillations was recorded. This was done 10 times to obtain the average period T_d .

$$T_d = \frac{t_{avg}}{10}$$

Since the disk can be considered as a solid cylinder, the moment of inertia I_d can be determined using equation 1. From these values, the directrix D^* of the wire can be obtained by:

$$D^* = I_d \cdot \left(\frac{2\pi}{T_d}\right)^2$$

Next, the same measurement was recorded for the wheel suspended from the same wire and the period of oscillation was calculated. Using equation 3, the moment of inertia of the wheel I_w was determined.

b. Physical Pendulum

For this second setup, the wheel was suspended as a physical pendulum by its cutting edge. The wheel was then tilted at an angle of approximately 10° before it was released and start oscillating. The time of 30 oscillations was recorded. This was done 10 times to obtain the average period T_w .

The moment of inertia of the wheel by this axis of rotation is calculated using equation 5.

The moment of inertia about its center of mass using Steiner theorem, can be calculated as:

$$I_w = I'_w - m \cdot s^2$$

5 Final Analysis

For the first measurement (the oscillating disk suspended from a wire) the time of 15 oscillations recorded are as follows:

	meas 1	meas 2	meas 3	meas 4	meas 5	meas 6	meas 7	meas 8	meas 9	meas 10	average <t>
time	44.16	44.08	44.18	44.12	43.95	43.99	44.16	44.11	44.04	44.36	44.12

(8)

Using the given average time, the period T_d

6 Summary

The results confirmed the expected physical relationships, predicted by theory.

Literature

Praktikum der Physik, Walcher, [Teubner Verlag 1989, Seite 83 ff]

Physik für Fachhochschulen und techn. Berufe, Heywang, Treiber, Herberg, Neft, [Handwerk und Technik, Hamburg 1992, Seite 200 ff]